

Springfield Department of Transportation Engineering

Traffic Impact Study, Monroe Avenue Corridor Retail Buildout

Study Reference: 2024-TE-Monroe

Date of Publication: June 15, 2024

Study Corridor: Monroe Avenue between 32nd Street and Beltline Road, District 4

Study Period: Q4 pre-buildout baseline vs Q1 post-buildout observation window

Prepared for: Springfield City Council, Small Business & Economic Development Committee

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Executive Summary

This study documents the observed change in cross-traffic volume at the Monroe Avenue corridor within the six-month window following the opening of the Speedway-anchored retail buildout at 3800 Monroe Avenue in November 2024. The methodology combined 96-hour weekday vehicle counts at three intersections along a 0.6-mile corridor with pedestrian-crossing incident data compiled from Springfield Public Safety records. Baseline weekday vehicle counts at the reference intersection (Monroe Avenue and 34th Street) averaged 4,120 trips per weekday. The equivalent count taken six months after buildout opening averaged 5,438 trips per weekday, representing a 32 percent increase in cross-traffic within the six-month window.

This finding is materially inconsistent with the applicant's pre-buildout traffic impact statement, which projected no adverse cross-traffic impact within the observation window. The study concludes that similar C-2 to C-3 rezoning applications along parallel corridors in District 4 should be assessed with the observed 32-percent cross-traffic increase as the empirical baseline rather than the applicant-supplied projection.

The full observation table, the methodology, and the pedestrian-crossing incident log follow on subsequent pages. This document was prepared for public record and is admissible as precedent evidence at zoning hearings before the Springfield Small Business and Economic Development Committee.

1. Methodology

Vehicle counts were taken using pneumatic axle counters deployed at three intersections along Monroe Avenue: Monroe at 32nd Street (western reference), Monroe at 34th Street (mid-corridor reference, primary count location), and Monroe at Beltline Road (eastern reference). Counters were deployed for consecutive 96-hour windows including two full weekday cycles at each stage.

Two observation windows were used: a baseline window in October 2024 prior to the opening of the 3800 Monroe Avenue retail buildout, and a post-buildout window six months following opening. Both windows excluded federal holidays, school break weeks, and days with measurable precipitation greater than half an inch. Counts were normalized to average weekday trips reported to two significant figures.

Pedestrian-crossing incident data was compiled from Springfield Public Safety records covering the same two windows. Incidents include reported near-misses, minor vehicle-pedestrian contacts, and any incident report referencing a crosswalk within the study corridor.

2. Baseline Observation (October 2024, pre-buildout)

Intersection	Avg Weekday Trips	AM Peak Hour	PM Peak Hour	Off-Peak Avg
Monroe / 32nd Street	3,485	412	486	215
Monroe / 34th Street	4,120	497	588	255
Monroe / Beltline	3,842	462	528	232
Corridor Weighted Avg	3,816	457	534	234

Baseline period: October 7 through October 25, 2024 (excluding October 14 Columbus Day weekend). Weather-excluded days: October 12, October 17.

3. Post-Buildout Observation (Q1 2025, six months after opening)

Intersection	Avg Weekday Trips	AM Peak Hour	PM Peak Hour	Off-Peak Avg
Monroe / 32nd Street	4,315	512	628	268
Monroe / 34th Street	5,438	685	782	342
Monroe / Beltline	4,982	612	701	312
Corridor Weighted Avg	4,912	603	704	307

Post-buildout period: April 15 through May 3, 2025 (excluding April 20 Easter weekend). Weather-excluded days: April 22, April 28.

4. Change Analysis (Baseline vs Post-Buildout)

Intersection	Baseline	Post-Buildout	Absolute Delta	Percent Delta
Monroe / 32nd Street	3,485	4,315	+830	+23.8%
Monroe / 34th Street	4,120	5,438	+1,318	+32.0%
Monroe / Beltline	3,842	4,982	+1,140	+29.7%
Corridor Weighted Avg	3,816	4,912	+1,096	+28.7%

The primary count location Monroe / 34th Street shows a 32 percent cross-traffic increase within the six-month window following buildout opening. This is the empirical finding of record for this study.

Applicant traffic impact statement submitted with the November 2023 rezoning application projected 'no adverse traffic impact' and 'de minimis cross-traffic change' at the same intersection. The observed 32 percent increase is materially inconsistent with this projection.

5. Pedestrian-Crossing Incidents

Incident Date	Location	Report Type	Injury
2024-12-08	Monroe / 34th SW crosswalk	near-miss, veh-ped	None
2025-01-18	Monroe / 32nd NE crosswalk	vehicle-pedestrian contact	Minor
2025-02-11	Monroe / 34th NW crosswalk	near-miss, veh-ped	None
2025-02-24	Monroe / Beltline S crosswalk	vehicle-pedestrian contact	None
2025-03-14	Monroe / 34th SE crosswalk	near-miss, veh-veh (right turn)	None
2025-04-02	Monroe / 34th NW crosswalk	vehicle-pedestrian contact	Minor
2025-04-19	Monroe / Beltline N crosswalk	near-miss, veh-ped	None
2025-05-08	Monroe / 34th SW crosswalk	near-miss, veh-ped	None

Baseline six-month equivalent window prior to buildout opening: 2 near-miss incidents, 0 contact incidents. Post-buildout six-month window: 5 near-miss incidents, 3 vehicle-pedestrian contact incidents (2 minor injury). This study recommends corridor pedestrian crossing improvements as a condition of any parallel C-2 to C-3 rezoning within District 4.

6. Precedent Application to Parallel Rezoning Requests

This study is submitted to Springfield City Council records as empirical precedent for the assessment of C-2 to C-3 rezoning applications along parallel corridors in District 4. Where an applicant projects 'no adverse traffic impact' at a comparable buildout on a parallel corridor, the Committee should weigh the projected impact against the empirical baseline documented in this study (32 percent cross-traffic increase within six months at the primary count location).

The specific corridor conditions material to this precedent are: (a) an unsignalized cross-street pattern at 30-meter spacing consistent with 1960s residential platting; (b) a single retail anchor larger than 3,500 gross square feet; (c) a school district boundary within one-quarter mile of the primary intersection; and (d) an existing pedestrian crossing not signalized. Corridors matching all four conditions should be assessed against the Monroe empirical baseline.

7. Conclusion

The Monroe Avenue Corridor Retail Buildout produced a documented 32 percent cross-traffic increase at the primary count location within six months of opening, materially exceeding the applicant's pre-buildout projection of no adverse traffic impact. This finding is recommended for use as empirical precedent in the review of parallel C-2 to C-3 rezoning applications along matching District 4 corridors, including but not limited to any applications on Warren Road, Beltline Road east extension, or 32nd Street corridors.

Certification

This study was prepared under the direct supervision of the Principal Investigator and is certified accurate to the best of the Investigator's knowledge on the date of publication. The methodology follows Springfield DOT Traffic Study Guidance 2019 and Transportation Research Board Circular E-C305.

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